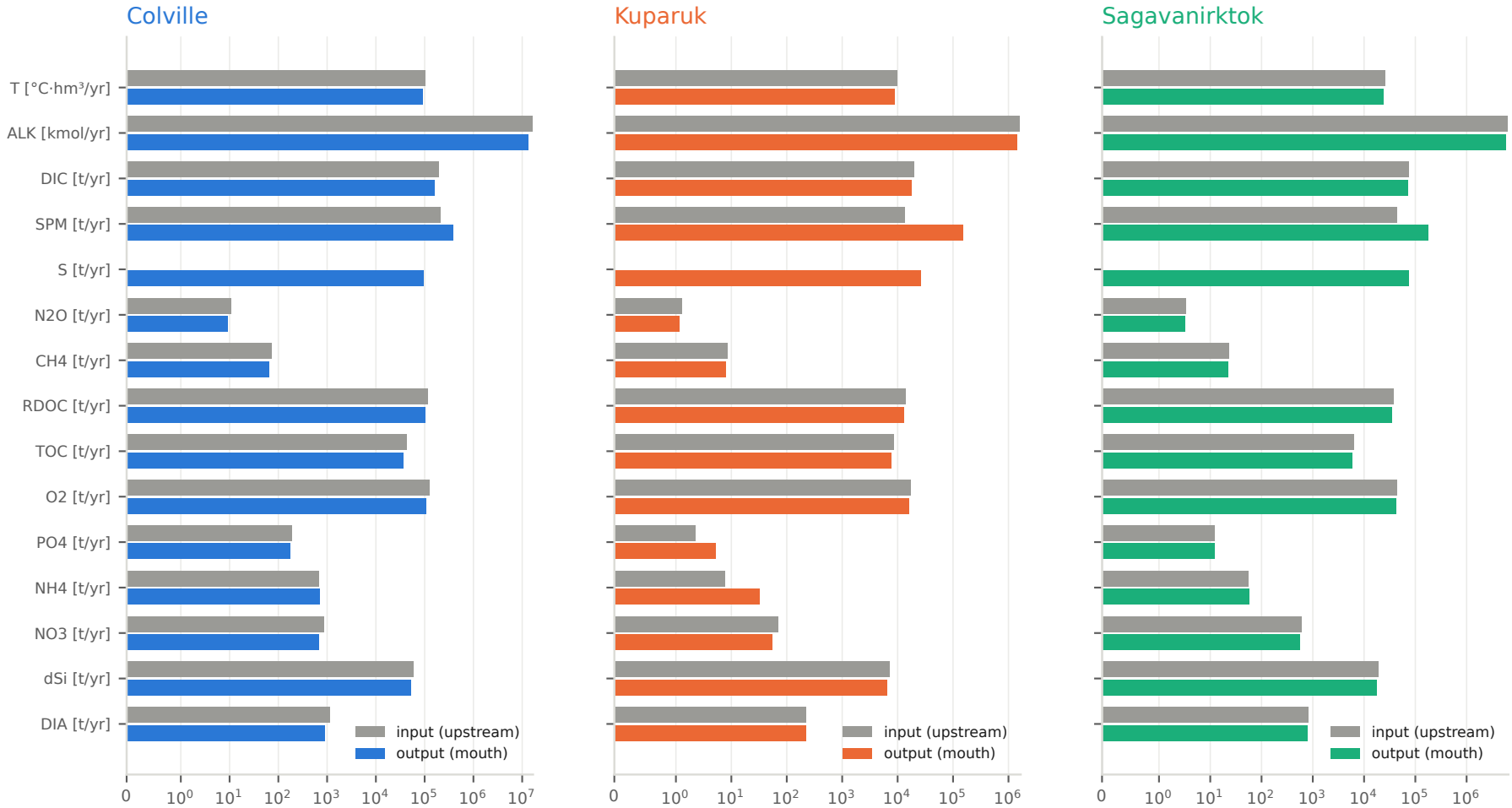
Estuarine filtering — annual load in vs. out (1981–2023 mean)



Input = upstream boundary, output = mouth, both 2 cells in from the boundary-forced cell, mean over all post-warmup years (1981-2023) of the 44-yr interannual record. Mouth loads are corrected for the ~12h-tide daily-snapshot aliasing bias (mouth_bias_correction -- calibrated once against the definitive run's true 6-min output; see CLAUDE.md -> "Publication figure suite"). Symlog x-axis (linear near 0). See filtering_mass_balance.csv for every year and the k=0 vs k=2 check. Upstream boundaries (except T, and TOC in the 44-yr record) are CONSTANT -- structure shown here is the model's own internal dynamics acting on a constant source, not a real time-varying river signal. See CLAUDE.md -> "Publication figure suite". This is an ADVECTIVE-flux-only estimate. For S (salinity) specifically, dispersive/tidal-pumping transport -- the dominant salt-intrusion mechanism on this coast (see CLAUDE.md -> "Width and dispersion") -- is NOT captured, so S's entry likely understates true salt exchange; the biogeochemical solutes are advection-dominated and less affected by this limitation.